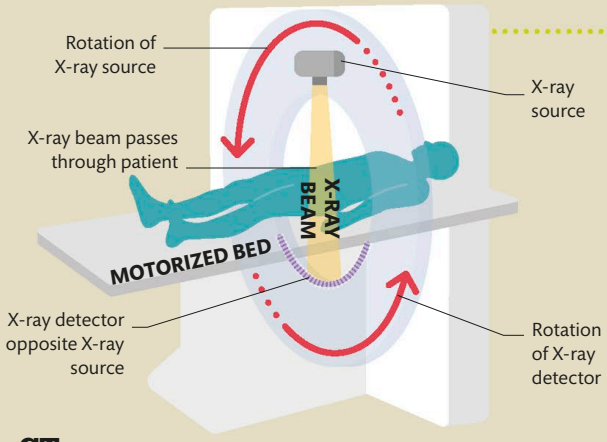


CT scans

Computer tomography (CT) scanning is a type of X-ray imaging that produces cross-sectional images ("slices") through the body. During a CT scan, the X-ray source and detector rotate around the patient, who lies on a motorized bed that moves forward with each scan. The detector is extremely sensitive, and the image data from it can be processed by computer to create highly detailed or even 3-D images of body tissues.

OTHER TYPES OF MEDICAL X-RAYS

In addition to plain X-rays and CT scans, there are various specialized X-rays, some of which use a contrast medium (a substance opaque to X-rays) to highlight specific tissues.

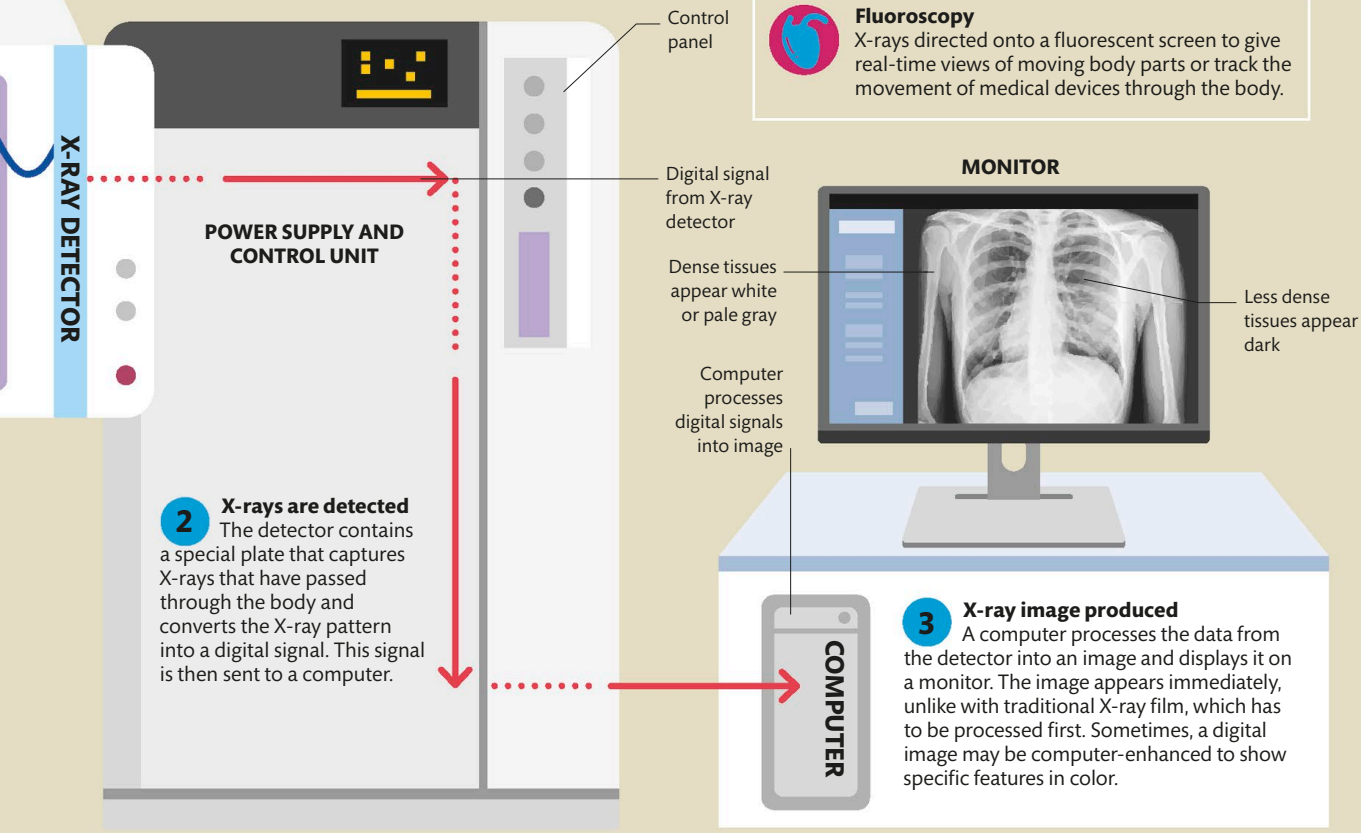
Dental X-rays
Low-dose X-rays of the teeth and jaws to reveal dental problems such as decay, abscesses, or disorders of the gums or jawbone.

Bone-density scanning
Low-dose X-ray scanning to reveal any areas of low bone density; usually carried out on the spine or pelvis to check for osteoporosis.

Mammography
Low-dose X-ray imaging of the breasts to detect any abnormalities such as tumors; frequently carried out to screen for breast cancer in women.

Angiography
X-ray imaging of the heart and blood vessels using an injected liquid contrast medium to show the interior of these structures clearly.

Fluoroscopy
X-rays directed onto a fluorescent screen to give real-time views of moving body parts or track the movement of medical devices through the body.



2 X-rays are detected
The detector contains a special plate that captures X-rays that have passed through the body and converts the X-ray pattern into a digital signal. This signal is then sent to a computer.

3 X-ray image produced
A computer processes the data from the detector into an image and displays it on a monitor. The image appears immediately, unlike with traditional X-ray film, which has to be processed first. Sometimes, a digital image may be computer-enhanced to show specific features in color.